Jeff Tan

CONTACT

Website: https://jefftan969.github.io
Email: jefftan1@stanford.edu

EDUCATION

Stanford University, Stanford, CA 09/2025 - Present
Ph.D. in Computer Science
- Advisor: Prof. Jiajun Wu

Carnegie Mellon University, Pittsburgh, PA 08/2023 - 08/2025
M.S. in Robotics (Research Thesis, GPA 4.17)
- Thesis: *Dynamic and Large-Scale 3D Reconstruction via Test-Time Optimization with Priors*
- Advisor: Prof. Deva Ramanan

Carnegie Mellon University, Pittsburgh, PA 08/2019 - 05/2023
B.S. in Computer Science (GPA 3.96)
- Thesis: *Distilling Neural Fields for Real-Time Articulated Shape Reconstruction*
- Concentration in Computer Graphics, Computer Systems, Algorithms

PUBLICATIONS

DyCraft: An Agentic System for Long-Horizon 4D Scene Generation.
Jeff Tan, Hang Dai, Zeqian Long, Yanzhe Lyu, R. Kenny Jones, Shangzhe Wu, Chen Geng, Jiajun Wu.
Under Review, 2027. [Website] [Paper] [Code]
- An agentic system for minute-long animated 4D scene generation with GPT-6-Astra.

Benchmarking Calibration-Robust Stereo Geometry Estimation in the Wild.
Jeff Tan, Nikhil Keetha, Yifei Liu, Shubham Tulsiani, Deva Ramanan.
CVPR-W, 2026. [Paper]
- We find that stereo geometry estimators perform poorly on real-world in-the-wild data.

CRISP: Contact-guided Real2Sim from Monocular Video with Planar Scene Primitives.
Zihan Wang*, Jiashun Wang*, **Jeff Tan**, Yiwen Zhao, Jessica Hodgins, Shubham Tulsiani, Deva Ramanan.
ICLR, 2026. [Website] [arXiv] [Code]
- We significantly improve terrain reconstruction for humanoid real-to-sim from RGB videos.

MonoFusion: Sparse-View 4D Reconstruction via Monocular Fusion.
Zihan Wang, **Jeff Tan**, Tarasha Khurana, Neehar Peri, Deva Ramanan.
ICCV, 2025. [Website] [arXiv] [Code]
- We reconstruct dynamic human-centric scenes from very sparse-view video.

DiffusionSfM: Predicting Structure and Motion via Ray Origin and Endpoint Diffusion.
Qitao Zhao, Amy Lin, **Jeff Tan**, Jason Y. Zhang, Deva Ramanan, Shubham Tulsiani.
CVPR, 2025. [Website] [arXiv] [Code]
- Pointmap estimation from sparse-view images via end-to-end diffusion.

DressRecon: Freeform 4D Human Reconstruction from Monocular Videos.
Jeff Tan, Donglai Xiang, Shubham Tulsiani, Deva Ramanan, Gengshan Yang.
3DV, 2025 (Oral). [Website] [arXiv] [Code]
- We reconstruct humans in loose clothing and interacting with objects, given a single monocular video as input, using a hierarchical deformation field and image-based priors.

Distilling Neural Fields for Real-Time Articulated Shape Reconstruction.
Jeff Tan, Gengshan Yang, and Deva Ramanan.
CVPR, 2023. [Website] [Paper] [Code]
- We learn real-time feed-forward pose and shape predictors, by distilling from dynamic NeRFs.

Awards

NSF Graduate Research Fellowship 2023 - 2028

CMU Alumni Award for Undergraduate Excellence 2023

CMU Summer Undergraduate Research Fellowship 2021

Research Experience

Google Research, San Francisco
Student Researcher (Host: Dr. Jason Zhang) 06/2026 - Present
- Content-adaptive discrete 1D image tokenization

Stanford University, Stanford Vision and Learning Lab 09/2025 - Present
Graduate Student Researcher (Advisor: Prof. Jiajun Wu)
- Real-to-sim human+object+scene reconstruction for humanoid-object interaction
- Agentic 4D animation and scene generation

Carnegie Mellon University, Center for Autonomous Vehicle Research 08/2023 - 08/2025
Graduate Student Researcher (Advisor: Prof. Deva Ramanan)
- Dynamic 4D reconstruction from monocular and sparse-view video
- Large-scale, photorealistic 3D site modeling from aerial and ground imagery (w/ IARPA WRIVA)
- Geometry and pointmap estimation via pretrained diffusion models

Carnegie Mellon University, Center for Autonomous Vehicle Research 05/2022 - 08/2023
Undergraduate Researcher (Advisor: Prof. Deva Ramanan)

Teaching Experience

Carnegie Mellon University, Pittsburgh, PA
- Teaching Assistant, Physics-Based Rendering (15-468) Spring 2023, Spring 2024
- Teaching Assistant, Parallel Computation (15-418) Fall 2021, Spring 2022, Spring 2023
- Teaching Assistant, Introduction to Computer Systems (15-213) Fall 2021

Service

Reviewer: CVPR, ECCV, ICLR, NeurIPS, WACV

Software

Lab4D: A framework for in-the-wild 4D reconstruction from monocular videos.
Gengshan Yang, **Jeff Tan**, Alex Lyons, Neehar Peri, Deva Ramanan.
[Code] [Docs]
A Python library for 4D reconstruction of humans, animals, and scenes from monocular videos.

Skills

Programming: Python, C++, C, OCaml, JavaScript, x86 Assembly

Software: PyTorch, JAX, NumPy, CUDA

Languages: English (native), Chinese (fluent)

Citizenship: United States